

DESK PILOT

CONTEXTUAL LANDSCAPE OF INTELLIGENT PRODUCT TRACEABILITY IN NIGERIA

PART REQUIREMENTS FOR SEMINAR

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NAME: [CLIENT NAME WITHHELD]

MATRIC NO: [REDACTED]

AOI: INTELLIGENT PRODUCT TRACEABILITY

YEAR: 2025/2026

FACILITATOR: [WITHHELD]

E-TUTOR(S): [WITHHELD]

Nigeria as the Research Context: Why It Matters

Africa's largest economy and digital trade champion, a nation at a critical crossroads of growth and supply chain vulnerability

- Nigeria stands as Africa's most populous nation, over 220 million people and its largest economy by GDP. It is the continent's acknowledged digital trade leader, having been endorsed by the African Union as the AfCFTA Digital Trade Champion in 2024 (CLG Global, 2025). The ICT sector contributed 19.78% to Nigeria's real GDP in Q2 2024, a figure that has grown steadily year on year (NBS, 2024), reflecting the speed at which digital infrastructure is reshaping the country's economic architecture. The digital transformation market, valued at \$9.58 billion in 2024, is projected to reach \$26.98 billion by 2030, a CAGR of 18.17%, driven by IoT adoption, AI investments, and government broadband expansion programmes (Mordor Intelligence, 2025).
- Yet behind these impressive headline numbers lies a stark contradiction: Nigeria is simultaneously one of the world's most vulnerable nations when it comes to supply chain integrity, product authenticity, and traceability. NAFDAC, the national food and drug regulator, has acknowledged that a significant percentage of imported medicines are substandard or counterfeit (NAFDAC, 2024). Food fraud, from adulterated grains and fake dairy products to counterfeit processed foods, has been described as an 'epidemic' in Nigerian urban life, affecting consumer confidence, public health, and business reputations across sectors (Njiru *et al.*, 2025). This paradox makes Nigeria an especially compelling and urgent setting for research into intelligent product traceability: the country has both the technological ambition and the regulatory imperative to build modern traceability systems, yet the academic evidence base for how these systems work in the Nigerian context remains remarkably thin.
- This desk pilot maps the Nigerian landscape across the five thematic gaps identified in the literature, blockchain technology, IoT integration, supply chain transparency, consumer trust and food safety, and organisational performance and sustainability, grounding the proposed research in the specific realities of Nigeria's industrial, regulatory, and technological environment.

Nigeria's Digital Economy at a Glance (2024-2025)

Key indicators underpinning the feasibility of intelligent traceability systems in Nigeria

19.78%

ICT GDP Contribution

Q2 2024 (NBS, 2024)

\$9.58B

Digital Market Value

2024 → \$26.98B by 2030

107M

Internet Users

January 2025 (DataReportal, 2025)

150M

Mobile Connections

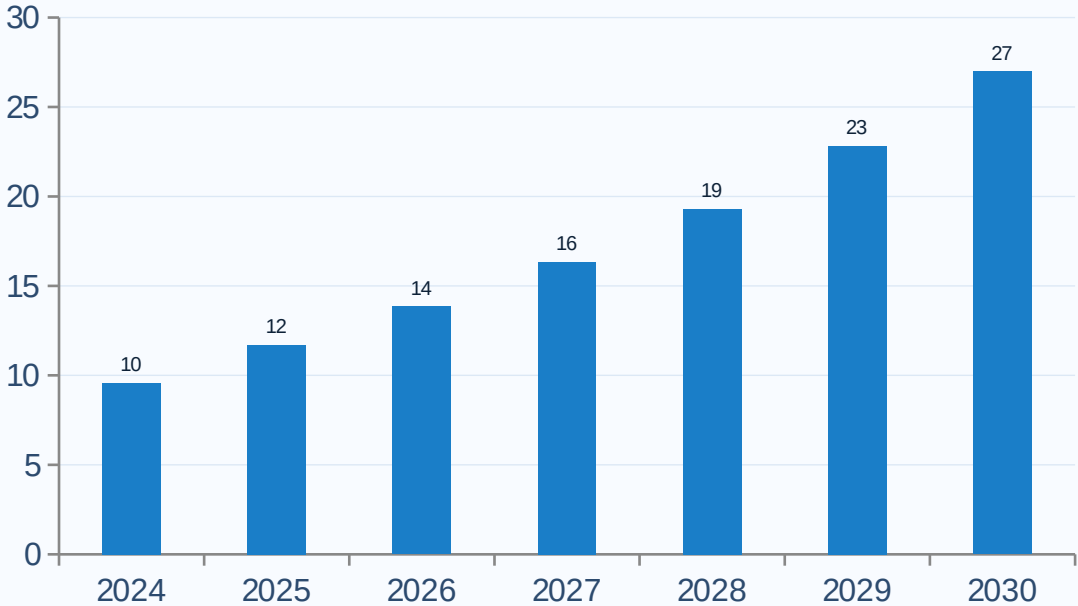
94.4% on broadband networks

48%

Broadband Penetration

2024 — 119M subscribers

Nigeria Digital Transformation Market (USD Billion) · Mordor Intelligence (2025)



Blockchain Technology in the Nigerian Context

Gap 1: Intelligent Product Traceability and Blockchain Technology

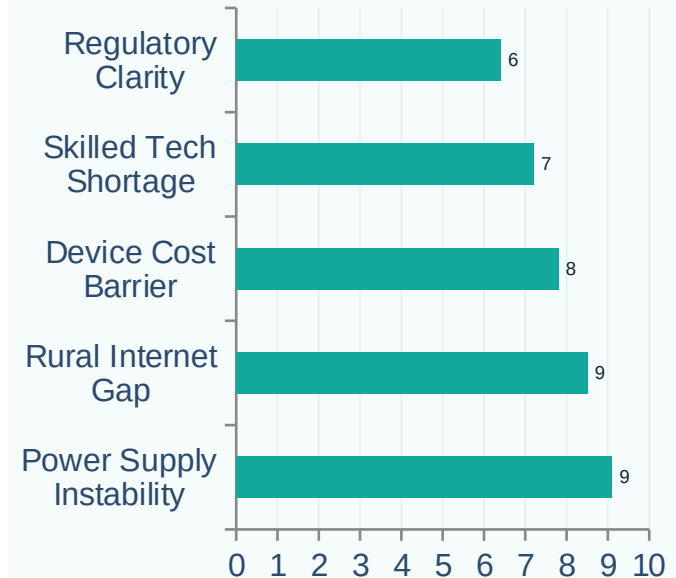
- Nigeria's engagement with blockchain technology has moved from conceptual enthusiasm to cautious regulatory action in recent years. In 2024, the Nigerian federal government announced plans to develop a sovereign blockchain infrastructure dubbed 'Nigerium,' intended to enhance national data security and reduce dependence on foreign-controlled platforms (Elkoraichi, Elfezazi, & Belhadi, 2025). Simultaneously, the Lagos State government announced the deployment of a blockchain-based system for tokenizing real estate transactions, a clear signal that sub-national governments are beginning to see distributed ledger technology as a practical governance tool rather than a futuristic aspiration. At the private sector level, Cellulant, a leading Nigerian payments company, has built 'AgriKore,' a blockchain-powered digital marketplace that links smallholder farmers directly to buyers, financial institutions, and supply chain participants across the agricultural value chain (Elkoraichi, Elfezazi, & Belhadi, 2025).
- Despite this momentum, the Nigerian blockchain landscape in supply chain and traceability applications remains in its infancy. The existing pilot projects, AgriKore, real estate tokenization, are primarily transactional and financial in orientation; they do not address the specific challenge of product-level traceability where every unit of a manufactured or agricultural good must carry a verifiable, tamper-proof digital identity from production to consumption. Akazue, Yoro and colleagues (2023) specifically developed a food value chain traceability system using blockchain technology in Nigeria, demonstrating the technical feasibility of the approach, yet their prototype was evaluated in a controlled, small-scale environment and has not been scaled to the broader manufacturing or pharmaceutical sectors. The infrastructure constraints that undermine blockchain deployment in Nigeria, unreliable internet connectivity, high energy costs affecting node operation, inadequate cloud adoption, and a shortage of blockchain developers, remain real structural barriers that global traceability literature has not adequately addressed.
- The regulatory environment for blockchain in Nigeria is also ambiguous. While NAFDAC enacted the Pharmaceutical Products Traceability Regulations in October 2024, mandating unique identifiers, data carriers, and mandatory reporting within 48 hours of any supply chain discrepancy (GS1 Nigeria, 2024), the regulations do not prescribe blockchain as the underlying technology. This creates an important research opportunity: determining whether blockchain's immutability and decentralisation advantages outweigh its scalability and cost constraints in the specific context of Nigeria's pharmaceutical and agri-food supply chains, where counterfeit products represent not just an economic problem but a documented public health emergency.

IoT Integration and Real-Time Traceability in Nigeria

Gap 2: Intelligent Product Traceability and IoT Integration

- Nigeria's IoT ecosystem is growing rapidly but unevenly. In July 2024, the IoT West Africa Conference and Exhibition was announced in Lagos, with the Association of Telecommunications Companies of Nigeria (ATCON) formally committing to accelerate IoT adoption across industries including agriculture, healthcare, and logistics (Mordor Intelligence, 2025). Hexing Nigeria's 2024 launch of advanced IoT solutions for smart metering demonstrates that the country's energy and utility sectors are beginning to invest seriously in sensor-based real-time monitoring infrastructure.
- However, for agricultural and manufacturing supply chains, where product traceability is most urgently needed, IoT adoption faces formidable barriers. Nigeria's rural internet penetration remains constrained; as of January 2025, over 54% of the population remained offline (DataReportal, 2025), meaning that IoT-enabled traceability devices in farms or rural processing facilities cannot rely on consistent connectivity. Power supply remains a chronic challenge, with RFID readers, GPS trackers, and environmental sensors all requiring stable electricity that is not reliably available outside major urban centres. The government's 'Fiber Forward' initiative (launched June 2024) to lay 90,000 kilometres of fibre optic infrastructure and the '774 LGA Connectivity' project to connect all local government secretariats to the internet represent critical enabling investments, but their full deployment and impact on rural supply chain connectivity will not be felt for several years.
- This creates a specific research gap that no paper in the reviewed literature addresses directly: how do you design IoT-based intelligent traceability systems that function reliably in low-connectivity, low-power, intermittently-online environments of the kind common across Nigeria's agricultural heartlands? The existing literature is almost exclusively designed for and validated in high-infrastructure environments, China, Europe, the United States, where stable internet and electricity are assumed to be available.

IoT Adoption Barriers in Nigeria
(Indicative Severity, Author's Assessment)



Supply Chain Transparency in the Nigerian Manufacturing & Agri-Food Sector

Gap 3: Intelligent Product Traceability and Supply Chain Transparency

- Nigeria's supply chains across agri-food, pharmaceuticals, manufacturing, and fast-moving consumer goods, are characterised by fragmentation, opacity, and weak multi-stakeholder coordination. The country operates a complex layered distribution system in which goods pass through multiple middlemen, aggregators, wholesalers, sub-wholesalers, and open market traders, before reaching consumers. Each handover point is a potential site of product adulteration, substitution, or documentation manipulation. The Food and Agriculture Organisation has long identified post-harvest losses across Nigeria's agricultural supply chain at between 30% and 50% of total produce, partly attributable to a lack of visibility and accountability across the supply chain. In food, pharmaceuticals, and electronics, the absence of real-time product tracking means that recalls are slow, contamination sources are difficult to identify, and consumers bear the cost of unsafe or substandard goods.
- Nigeria's recognition as the African Union's AfCFTA Digital Trade Champion in 2024 carries specific implications for supply chain transparency. The AfCFTA Digital Trade Protocol, adopted in February 2024, includes provisions on data transfers, digital identities, and cross-border supply chain management, all of which have direct relevance to traceability. Nigeria has signed the African Union's Convention on Cybersecurity and Personal Data Protection (January 2024) and has enacted the Nigeria Data Protection Act, establishing legal frameworks that govern how supply chain data is collected, stored, and shared across borders. Yet the practical capacity of Nigerian supply chain actors, particularly smallholder farmers, small and medium enterprises, and informal market traders who represent the bulk of the distribution system, to participate in and benefit from digital traceability platforms has received minimal empirical investigation.
- The government has recognised supply chain transparency as a national priority in specific sectors. In the pharmaceutical sector, NAFDAC's Pharmaceutical Products Traceability Regulations 2024 (enacted October 30, 2024) require all manufacturers, importers, and distributors to assign unique identifiers to each packaging level and share transactional data with the NAFDAC national database, a mandate that covers the entire pharmaceutical supply chain from factory to pharmacy (GS1 Nigeria, 2024). In the food sector, the NSPRI Food Safety Policy (2024–2030) established a seven-year framework to modernise postharvest management and supply chain oversight. These regulatory commitments create a compelling national context for empirical research into whether and how intelligent traceability systems can operationalise the transparency that Nigeria's regulators are now legally requiring.

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Key Nigerian Regulatory Milestones in Product Traceability (2019–2025)

Nigeria's traceability regulatory timeline, the foundation for this research's relevance

YEAR	MILESTONE	IMPLICATION FOR TRACEABILITY RESEARCH
2019	NAFDAC convenes first national stakeholder summit on pharmaceutical supply chain traceability	Established the vision and roadmap; signalled government commitment to end-to-end traceability
2020	Nigeria National Traceability Strategy Policy published (Federal Ministry of Health)	Provided national policy framework and 5-year implementation targets for supply chain visibility
2024	NAFDAC Pharmaceutical Products Traceability Regulations 2024 enacted (October 30)	Mandatory unique identifiers, data carriers, and 48-hour error reporting for all pharmaceutical supply chain actors
2024	NSPRI Food Safety Policy (2024–2030) launched for postharvest supply chain management	Seven-year framework creating audit and traceability obligations across Nigeria's food supply chain
2024	Nigeria designated AfCFTA Digital Trade Champion; AfCFTA Digital Trade Protocol adopted (Feb)	Legal groundwork for cross-border digital identity, supply chain data sharing, and traceability interoperability
2024	'Nigerium' sovereign blockchain initiative announced; Lagos deploys blockchain for real estate	Government-level blockchain adoption signals intent to build national distributed ledger infrastructure

Consumer Trust, Food Safety and Product Authenticity in Nigeria

Gap 4: Intelligent Product Traceability and Consumer Trust / Food Safety

- Nigeria's consumer trust crisis in product safety is not theoretical, it is lived, documented, and deadly. NAFDAC has acknowledged that between 13% and 15% of medicines circulating in Nigeria are substandard or counterfeit; independent researchers and industry stakeholders have placed the real figure significantly higher, with NAFDAC admitting at one point that approximately half of the country's imported medicines, which account for over 70% of total consumption, may be fake (Think Global Health, 2024). The highest number of food fraud cases in sub-Saharan Africa in 2022 was recorded in Nigeria, particularly in grain-based cereals, a finding from peer-reviewed analysis published in Frontiers in Food Science and Technology (2025)(Njiru et al., 2025). Nigerian consumers are broadly aware of these risks: surveys document familiarity with high-profile food fraud cases including 'plastic rice' and counterfeit dairy products that gained national media attention.
- The economic and public health toll of these failures is staggering. The Grocery Manufacturers Association estimates food fraud inflicts \$10 to \$15 billion in damage annually on the global food industry, with approximately 10% of all commercially sold food products worldwide affected and Africa's vulnerability is disproportionately higher (Heliyon, 2024). A deeply tragic incident near Nigeria's capital, in which young lives were lost due to food adulteration, was cited in a 2024 peer-reviewed study as a stark illustration of the life-or-death stakes of food supply chain integrity in Nigeria.
- Yet Nigerian consumer behaviour research on traceability systems remains almost entirely absent. No published study has empirically tested how Nigerian consumers respond to blockchain-enabled product traceability labels, QR-code verification, or IoT-monitored supply chains, whether they understand what the technology means, whether it affects their purchase decisions, and whether they are willing to pay a premium for traceable products. The NSPRI Food Safety Policy (2024–2030) acknowledges this gap implicitly, noting that consumer food safety literacy and trust-building must accompany any technical traceability system. This is a significant research opportunity that this study is positioned to address.

13–15%Medicines counterfeit
(NAFDAC admission)**50%+**

Imported drugs estimated fake

#1

Nigeria: highest food fraud in sub-Saharan Africa (2022)

\$10–15B

Global annual cost of food fraud (GMA estimate)

Organisational Performance, Sustainability and Traceability in Nigeria

Gap 5: Intelligent Product Traceability and Organisational Performance / Sustainability

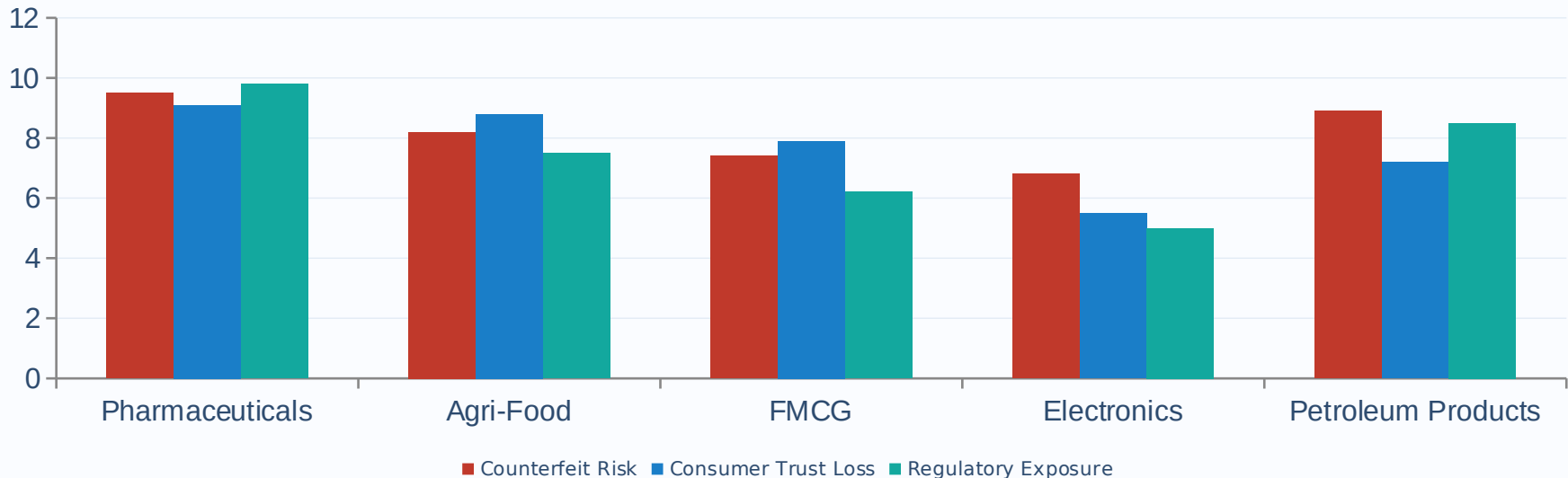
- For Nigerian businesses, intelligent product traceability is beginning to transition from a compliance obligation to a strategic competitive asset, though this transition is uneven and sector-specific. In the pharmaceutical sector, the mandatory NAFDAC traceability regulations create a direct business case: non-compliance carries criminal penalties and fines up to ₦5,000,000 for corporate entities, with the potential for asset forfeiture (GS1 Nigeria, 2024). Compliance is thus no longer optional, and the question for Nigerian pharmaceutical companies is not whether to implement traceability but how to do so efficiently, cost-effectively, and in ways that generate organisational value beyond regulatory adherence. In the agri-food sector, the pressure comes from both domestic regulation, the NSPRI Food Safety Policy 2024–2030 and export market requirements, as Nigeria's ambitions under the AfCFTA to expand agricultural exports to other African nations will require demonstrable traceability standards that meet continental expectations.
- The sustainability dimension of product traceability in Nigeria is especially underexplored. Nigeria loses an estimated 30% to 50% of agricultural produce post-harvest annually, a figure representing billions of dollars in wasted food, wasted inputs, and wasted carbon. Intelligent traceability systems that provide real-time visibility into product condition, location, and handling can directly reduce post-harvest losses by enabling faster interventions when environmental conditions deviate from safe parameters. This aligns traceability investment with SDG 12 (Responsible Consumption and Production), SDG 2 (Zero Hunger), and SDG 13 (Climate Action), sustainable development goals to which the NSPRI Food Safety Policy explicitly refers. However, no Nigerian study has quantified the sustainability return on investment from traceability system deployment, leaving this connection theoretical rather than empirically established.
- The competitive advantage dimension is equally important. As Africa's designated digital trade champion, Nigeria's ability to position its exports as verifiably authentic, safely produced, and sustainably sourced will become a significant determinant of market access under the AfCFTA digital trade framework. Companies that invest in blockchain-IoT traceability now, rather than waiting for mandatory compliance, will be positioned as preferred suppliers in regional and international markets where traceability is increasingly a procurement prerequisite. A 2024 study of Turkish manufacturing SMEs found that blockchain-enabled supply chain traceability positively influences sustainable supply chain practices (Abobassier et al., 2026), a finding that is highly relevant to Nigeria's own manufacturing SME sector but has never been empirically tested in the Nigerian context.

Nigeria's Traceability Gap: The Cost of Non-Traceability Across Key Sectors

Evidence base for why intelligent product traceability is a national priority in Nigeria

- Across Nigeria's five most critical product sectors , pharmaceuticals, agri-food, electronics, fast-moving consumer goods, and petroleum, the absence of robust intelligent traceability systems translates into quantifiable losses in revenue, public health outcomes, and organisational credibility. The chart below maps estimated impact severity across these dimensions, drawing on available regulatory, academic, and industry data.

Impact Severity Score (1–10) by Sector — Author's Assessment Based on NAFDAC, NSPRI, Njiru et al., (2025) & NBS (2024) Data



The Nigerian Research Context: Summary Assessment

01 Digital Ambition Is Real

Nigeria is Africa's digital trade champion with a \$9.58B transformation market growing at 18% CAGR, the infrastructure intent is there.

02 Traceability Need Is Urgent

Counterfeit drugs (13–50%), the highest food fraud rate in sub-Saharan Africa, and \$10–15B in annual global food fraud costs make traceability a life-or-death priority.

03 ➤ Regulation Is Ahead of Research

NAFDAC's 2024 Traceability Regulations and NSPRI's 2024–2030 Food Safety Policy mandate traceability, but the academic evidence base for Nigerian-context implementation is virtually absent.